

Supplementary material

S1. Method and evaluation details

The class-versus-rest standardized effect used for marker nomination is defined in the main text. The remaining equations specify the supplementary implementation details for projection, auxiliary losses, reliability calibration and evaluation metrics.

For the clustering representation, the pooled encoder representation $\mathbf{h}_i$ is mapped by the projection head $\mathcal{P}_\psi$ and L2-normalized:

$$\mathbf{z}_i = \frac{\mathcal{P}_\psi(\mathbf{h}_i)}{\|\mathcal{P}_\psi(\mathbf{h}_i)\|_2}. \quad (\text{S1})$$

For expression reconstruction, let Ω_i denote the reconstruction positions in profile i , and let $\hat{x}_{ig}$ be the predicted expression value. The reconstruction loss is

$$\mathcal{L}_{\text{expr}} = \frac{1}{\sum_i |\Omega_i|} \sum_i \sum_{g \in \Omega_i} (\hat{x}_{ig} - x_{ig})^2. \quad (\text{S2})$$

For supervised classification, the classifier predicts class probabilities p_{ic} from the pooled representation, and cross-entropy is computed on the labelled subset:

$$\mathcal{L}_{\text{ce}} = -\frac{1}{|\mathcal{L}|} \sum_{i \in \mathcal{L}} \log p_{i, y_i}. \quad (\text{S3})$$

For cross-view consistency, let $\mathbf{z}_i^{(1)}$ and $\mathbf{z}_i^{(2)}$ denote normalized embeddings from the original and perturbed views of profile i . The consistency loss is

$$\mathcal{L}_{\text{cons}} = \frac{1}{N} \sum_{i=1}^N \left[1 - \cos(\mathbf{z}_i^{(1)}, \mathbf{z}_i^{(2)}) \right]. \quad (\text{S4})$$

For module reliability, Qwen first fixes module membership and direction, after which reliability is estimated from the target dataset. Let e_c denote the exposed-label class-versus-rest effect of module c , t_c the fraction of exposed target-class profiles whose largest prior-assignment probability corresponds to module c , m_c the corresponding mean top-two assignment margin, and b_c the mean evidence score of Qwen-selected genes in the module. Only exposed labelled profiles are used to compute e_c , t_c and m_c ; unexposed labels are not used in reliability estimation. The implementation maps these quantities to bounded gates,

$$\begin{aligned} g_c^{\text{eff}} &= \sigma((e_c - \tau_{\text{eff}})/T_{\text{rel}}), \\ g_c^{\text{top}} &= \text{clip}(t_c, 0, 1), \\ g_c^{\text{mar}} &= \text{clip}(m_c/M_{\text{mar}}, 0, 1), \\ g_c^{\text{bio}} &= \text{clip}(\log(1 + b_c)/\log(1 + B_{\text{bio}}), 0, 1), \end{aligned} \quad (\text{S5})$$

and combines them as

$$r_c = \text{clip} \left[g_c^{\text{eff}} (0.45 + 0.35g_c^{\text{top}} + 0.20g_c^{\text{mar}}) (0.30 + 0.70g_c^{\text{bio}}), 0, 1 \right]. \quad (\text{S6})$$

If $e_c < \tau_{\text{eff}}$, r_c is multiplied by a low-effect shrinkage factor; if the resulting value is below the minimum reliability threshold, it is set to zero.

Together with the prototype and knowledge-alignment terms defined in the main manuscript, these objectives separate the sources of supervision used during adaptation. Reconstruction, consistency regularization and knowledge alignment are evaluated on labelled and unlabelled profiles, whereas cross-entropy and prototype learning use the labelled subset. The language-model output defines marker membership and direction, but continuous activity targets are computed from the expression data and reliability scores are calibrated using target-data effects, exposed-label agreement and biological evidence.

All clustering metrics were computed on all profiles in each benchmark dataset. Let y_i and $\hat{y}_i$ denote the reference label and predicted cluster assignment for profile i , respectively, and let π denote the optimal permutation between predicted clusters and reference labels. Clustering accuracy was calculated as

$$\text{ACC} = \frac{1}{N} \max_{\pi} \sum_{i=1}^N \mathbb{I}(y_i = \pi(\hat{y}_i)). \quad (\text{S7})$$

Normalized mutual information was defined as

$$\text{NMI} = \frac{2I(Y; \hat{Y})}{H(Y) + H(\hat{Y})}. \quad (\text{S8})$$

For adjusted Rand index, let n_{kj} be the number of profiles assigned to predicted cluster k and reference class j , and let a_k and b_j be the corresponding marginal totals. ARI was computed as

$$\text{ARI} = \frac{\sum_{k,j} \binom{n_{kj}}{2} - \frac{(\sum_k \binom{a_k}{2})(\sum_j \binom{b_j}{2})}{\binom{N}{2}}}{\frac{1}{2} \left[\sum_k \binom{a_k}{2} + \sum_j \binom{b_j}{2} \right] - \frac{(\sum_k \binom{a_k}{2})(\sum_j \binom{b_j}{2})}{\binom{N}{2}}}. \quad (\text{S9})$$

Purity was calculated as

$$\text{PUR} = \frac{1}{N} \sum_k \max_j n_{kj}. \quad (\text{S10})$$

Table S1. Categories of compared methods.

Category	Method	Reference
Semi-supervised learning	SSSE	Zeng et al. 2025
	scSemiPLC	Ma et al. 2026
	HiCat	Bi et al. 2025
	scANVI	Xu et al. 2021
	Mean Teacher	Tarvainen and Valpola 2017
	SSNMDI	Qiu et al. 2023
Unsupervised clustering	scMSCF	Jiang et al. 2025
	scMIB	Yan et al. 2026
	SEAL	Ye et al. 2026
LLM-derived gene representation	GenePT	Chen and Zou 2025
Biological foundation models	CellVQ	Wang et al. 2026
	RegFormer	Hu et al. 2026

Comparator evaluation protocol. For every benchmark, all compared methods used the same processed expression matrix, retained gene set, profile universe and number of classes K . For semi-supervised comparisons, a class-stratified 20% subset of profiles was exposed as labelled examples and the same exposed-label split was reused across applicable methods; reference labels from the remaining profiles were not supplied during fitting or hyperparameter selection. Methods requiring a pre-specified number of groups received the number of classes K , including the final k-means step of STCKLA. Comparator-specific settings followed the respective authors' documented recommendations or were selected using exposed labels only. The reported benchmark values correspond to the recorded stratified split with seed 0 and therefore represent fixed-split estimates, not multi-seed means.

The full numerical comparisons underlying the main-text benchmark figure are reported in Supplementary Tables S2 and S3.

Table S2. ACC and NMI (%) on benchmark datasets. Best values are bold.

Model	BRCA		Pan-cancer		4k PBMC		Lung		Liver		Pancreas		Melanoma	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
SSSE	46.96	33.91	76.75	83.84	40.95	28.76	38.92	19.04	41.97	24.55	43.13	40.31	56.00	37.20
scSemiPLC	69.51	38.33	93.98	92.43	83.56	79.11	85.88	72.46	91.28	87.22	97.19	92.01	87.38	75.39
HiCat	75.40	48.92	89.33	88.65	83.98	79.38	84.59	73.99	81.75	82.71	96.01	90.23	86.39	70.60
scMSCF	55.89	39.17	81.26	84.03	49.97	49.09	50.14	43.00	55.55	68.15	46.44	56.02	40.69	2.86
scANVI	55.74	37.70	79.81	86.69	54.12	60.04	44.78	45.48	38.71	49.61	26.30	35.77	30.85	30.76
Mean Teacher	77.35	58.54	94.30	93.54	57.89	68.14	63.10	64.92	74.28	79.55	79.86	81.95	69.90	61.48
SSNMDI	42.03	29.01	45.59	60.47	69.91	70.16	50.28	39.81	70.51	75.38	25.91	16.31	45.34	34.74
scMIB	41.13	11.49	77.44	83.67	73.86	74.65	39.10	51.62	62.66	73.12	44.62	47.37	47.97	56.56
SEAL	50.22	19.88	63.94	73.66	60.39	64.56	61.28	58.97	56.87	71.76	48.02	41.00	71.11	57.83
CellVQ	38.00	15.92	62.98	76.41	72.87	77.02	54.15	60.28	65.91	78.25	46.62	48.95	58.54	62.19
GenePT	38.00	3.78	22.23	36.73	37.21	51.23	41.30	36.56	56.28	62.30	23.56	28.52	41.25	39.20
RegFormer	45.28	21.80	43.36	57.53	60.14	68.87	51.56	61.60	61.01	70.20	35.52	43.64	58.48	63.18
STCKLA	89.42	73.30	95.65	94.26	89.62	85.91	90.60	81.09	95.39	92.24	98.22	94.59	95.05	87.47

Table S3. ARI and PUR (%) on benchmark datasets. Best values are bold.

Model	BRCA		Pan-cancer		4k PBMC		Lung		Liver		Pancreas		Melanoma	
	ARI	PUR	ARI	PUR	ARI	PUR	ARI	PUR	ARI	PUR	ARI	PUR	ARI	PUR
SSSE	20.50	67.70	71.34	88.12	19.09	41.47	15.13	39.76	23.76	42.18	19.80	61.05	32.64	69.14
scSemiPLC	43.04	71.85	91.41	94.03	72.88	83.56	75.99	85.88	82.75	91.28	95.70	97.20	81.11	88.39
HiCat	41.27	75.56	84.59	89.58	75.15	84.05	74.89	84.72	73.14	84.25	95.79	96.95	83.27	87.73
scMSCF	25.65	70.94	73.66	85.99	32.91	51.21	30.33	54.10	44.27	57.38	39.14	79.89	1.20	43.92
scANVI	29.39	71.14	74.41	88.48	40.04	68.06	31.30	62.74	26.55	44.80	14.49	55.80	19.65	69.13
Mean Teacher	66.27	83.61	92.91	94.71	49.79	70.53	48.14	77.92	66.82	82.00	71.41	94.92	66.03	83.27
SSNMDI	22.16	66.62	38.50	57.67	57.17	75.43	34.68	52.28	63.38	75.04	9.75	48.48	27.06	69.95
scMIB	9.42	56.93	74.17	85.12	64.05	82.92	29.35	60.12	52.50	71.15	31.54	70.65	39.35	84.16
SEAL	13.17	59.61	57.65	65.50	53.74	60.48	52.38	61.85	50.25	57.03	25.60	48.74	57.83	78.88
CellVQ	11.07	57.08	58.11	76.46	62.03	86.37	40.39	72.17	57.18	76.28	31.12	73.28	48.92	86.42
GenePT	2.80	53.80	9.56	29.10	29.80	60.41	25.26	52.15	43.51	65.86	11.47	52.41	26.07	72.53
RegFormer	14.16	60.92	35.73	54.43	49.65	74.00	41.03	71.07	46.59	66.62	19.19	62.62	47.98	87.06
STCKLA	72.33	89.42	93.41	95.65	83.64	91.56	81.97	90.60	89.73	95.39	96.57	98.22	92.83	95.05

S2. BRCA embeddings

Supplementary Figure S1 shows BRCA comparator embeddings under the same visualization setting.

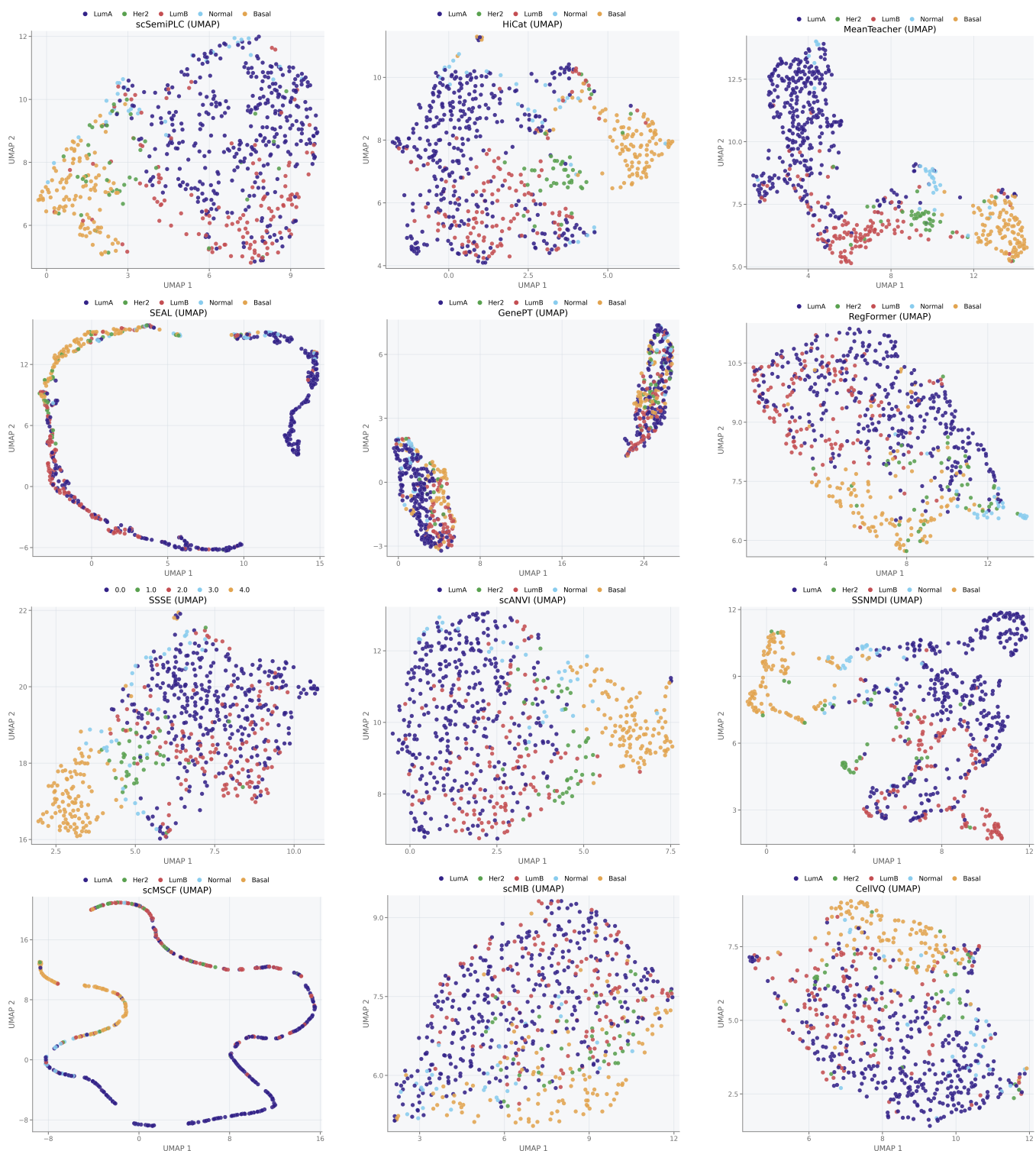


Figure S1 BRCA comparator UMAP visualizations. Panels show scSemiPLC, HiCat, Mean Teacher, SEAL, GenePT, RegFormer, SSSE, scANVI, SSNMDI, scMSCF, scMIB and CellVQ under the same visualization setting. Points are coloured by reference molecular subtype.

S3. Module validation

We examined the BRCA Qwen-assisted marker modules to determine whether their molecular organization remained specific to the target classes. Supplementary Table S4 provides the complete positive and negative BRCA marker programmes from the Qwen-assisted construction record. Supplementary Figure S2 displays the candidate-gene evidence and the resulting signed weights of the Qwen-organized modules. The controlled module-validation results, contribution enrichment, layer-wise alignment and module-activity distributions are reported in the corresponding main-text figures.

Table S4. Complete BRCA marker programmes from the Qwen-assisted construction record. Positive and negative genes are listed in final programme order.

Subtype	Positive markers	Negative markers
LumA	<i>ESR1, GATA3, IKBKB, FRS2, MAPT, SPEF2, DACH1, KANSL3, KLHDC1, CELSR1</i>	<i>TUBA1C, UBE2C</i>
Her2	<i>PSMD1, AKT1, MYCN, MMP1, EPCAM, G6PD, IDH1, MPHOSPH6, KMO, IDH2</i>	<i>ESR1, IGF1R, KRT14, RARG</i>
LumB	<i>SKP1, TUBA4B, UBE2N, RBBP7, YWHAB, ESR1, GATA3, BUB3, NDUFAF6, PCLAF</i>	<i>EGFR, SAA1, PLCG2, KRT14, DEFB1</i>
Normal	<i>EGFR, GNG7, TXNIP, JAG1, KIT, NLRP3, DLL1, NGFR, KRT14, DEFB1</i>	<i>RFC2, RMI1, RMI2, ACTL6A, CDC25C, TOMM70, RUVBL1, MCM2, CENPF, FOXM1</i>
Basal	<i>LYN, RIPK2, CDC20, AURKB, ACTL6A, LBR, E2F3, LPIN1, RUVBL1, CDCA8, MCM2, CENPF, POLR1C, LDHB, FOXM1, CENPN, SNRPA1, DEPDC1B, NDC80, NFYC, ARHGEF4, ORC1, PIM1, RAD51AP1, ORC6, SUMO3, XRCC2, COL9A3, RCC2, SPD11, DIAPH3, TYMS, RBM17, CDC45, PHF5A, TOP2A, PTTG1</i>	<i>GATA3, ELOVL5, TGFB3, SCNN1A, NEIL1, GAMT, PEX11A, SLC44A4, NAT1, DUSP5</i>

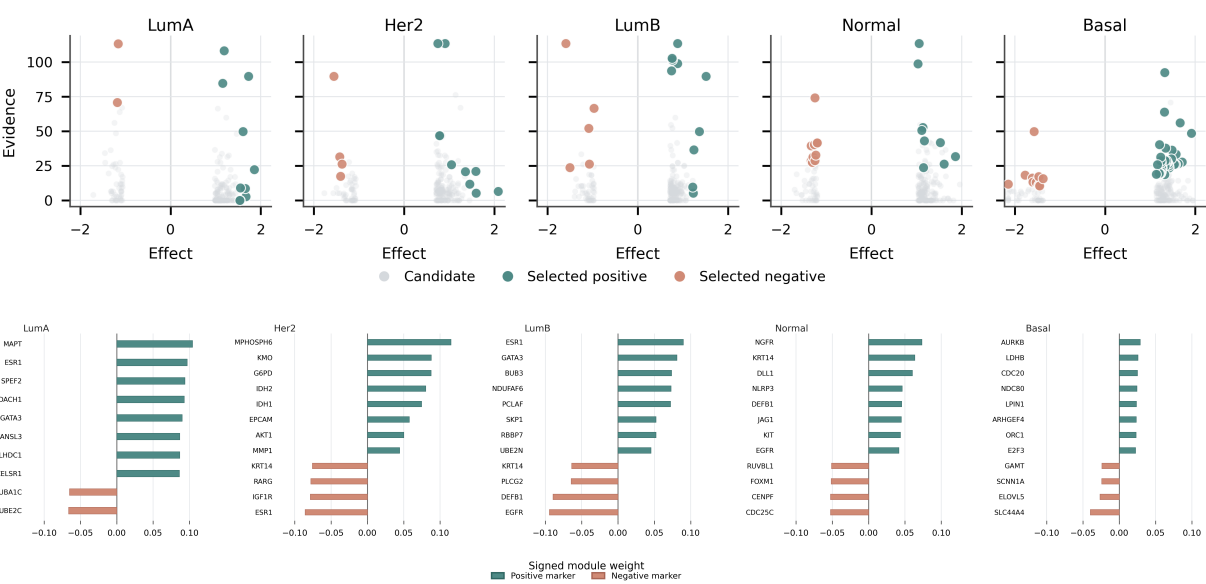


Figure S2 Qwen-assisted construction of BRCA class-specific marker modules. The upper panel shows candidate-gene effect and evidence values; the lower panel shows signed weights for selected positive and negative markers.

For the profile-specific activity control, the Qwen-derived class-specific modules, signed gene weights, module reliability coefficients, input capacity and training objectives were retained, but the activity vector of every profile was replaced by the dataset-level mean activity vector. The corresponding numerical results are reported in the main manuscript. This control isolates the contribution of continuous expression-profile-specific activity from a fixed dataset-level prior.

We further compared the class-specific module construction with a global-module control. The global control used the same candidate-gene budget and evidence sources, but organized genes without class-specific Qwen reasoning. Its numerical results are reported in the main manuscript; this control evaluates the contribution of class-resolved marker-module organization.

S4. Ablation results

The numerical BRCA ablation values are reported in the main-text ablation table. Supplementary Figure S3 provides the corresponding bar-plot visualization.

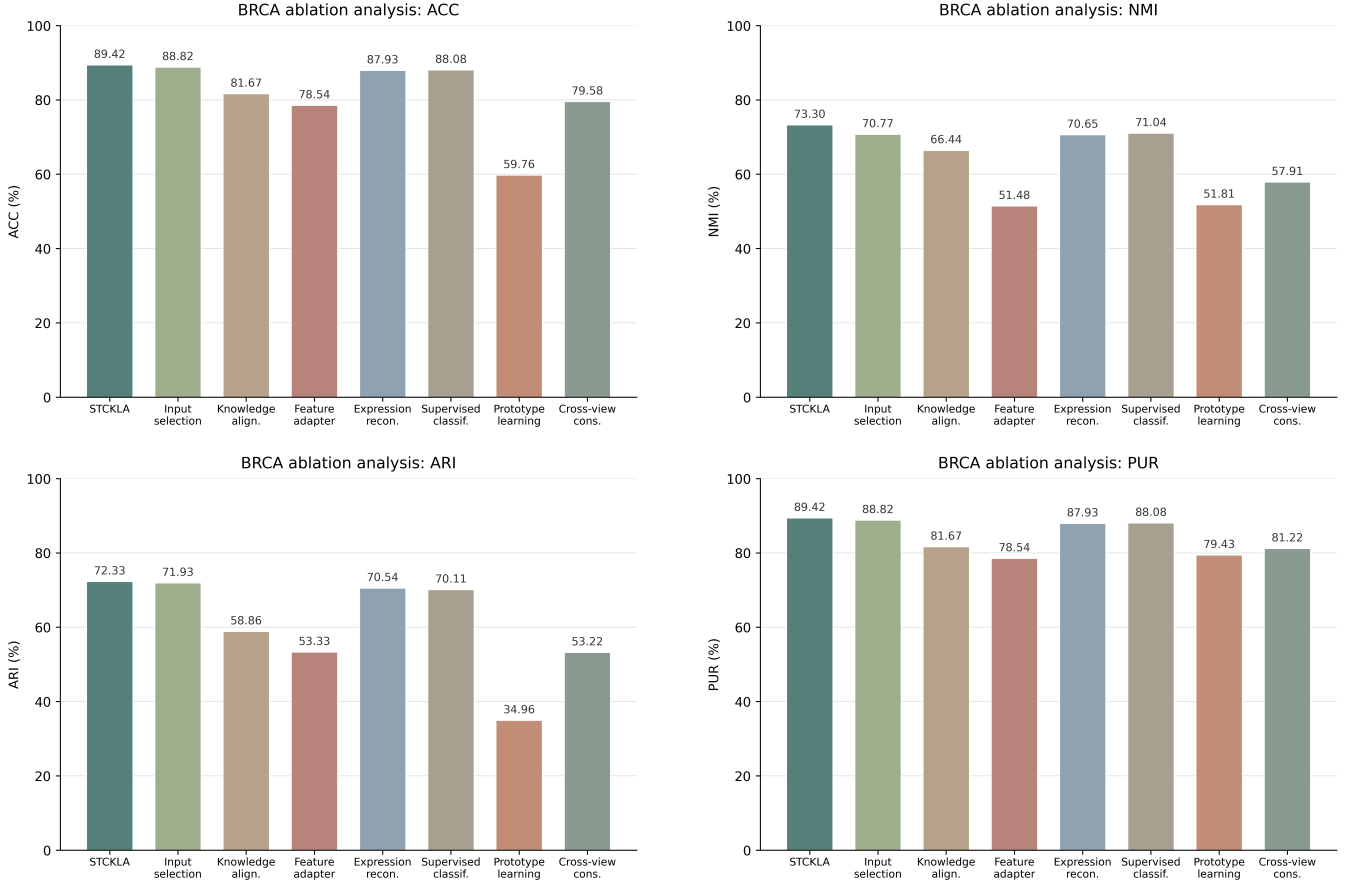


Figure S3 BRCA ablation analysis. Panels show ACC, NMI, ARI and PUR for STCKLA and one-component removal variants.

S5. Hyperparameter analysis

We performed one-factor sweeps on BRCA to evaluate the loss weights associated with knowledge alignment, prototype learning, expression reconstruction, supervised classification and consistency regularization. In the main manuscript, reconstruction, supervised classification and prototype learning are written compactly under the shared coefficient λ_{rep} , because the default benchmark configuration assigns the same weight to these three primary representation-learning terms. In the implementation and sensitivity analysis, their individual coefficients are retained as λ_{expr} , λ_{ce} and λ_{proto} . For each sweep, one implementation-level coefficient was varied around the default configuration while all other settings were fixed. The default setting was $\lambda_{\text{expr}} = \lambda_{\text{ce}} = \lambda_{\text{proto}} = 0.50$, $\lambda_{\text{cons}} = 0.10$ and $\lambda_{\text{know}} = 0.005$, corresponding to $\lambda_{\text{rep}} = 0.50$ in the grouped main-text objective.

Table S5. One-factor hyperparameter analysis on BRCA.

Hyperparameter	Value	ACC	NMI	ARI	PUR
λ_{know}	0.003	88.67	70.38	71.06	88.67
	0.005	89.42	73.30	72.33	89.42
	0.008	88.23	70.01	70.55	88.23
λ_{proto}	0.30	89.27	72.20	72.25	89.27
	0.50	89.42	73.30	72.33	89.42
	0.70	89.12	71.30	71.55	89.12
λ_{expr}	0.30	87.48	67.82	70.23	87.48
	0.50	89.42	73.30	72.33	89.42
	0.70	88.67	70.60	71.18	88.67
λ_{ce}	0.30	89.12	71.15	71.61	89.12
	0.50	89.42	73.30	72.33	89.42
	0.70	88.23	70.77	70.17	88.23
λ_{cons}	0.05	88.97	71.58	71.79	88.97
	0.10	89.42	73.30	72.33	89.42
	0.20	87.78	68.09	70.66	87.78

The default setting achieved the highest ACC, NMI, ARI and PUR in each one-factor sweep (Table S5; Figure S4). Decreasing or increasing the weight of any single implementation-level component reduced performance, supporting the fixed default configuration used for the main benchmark experiments.

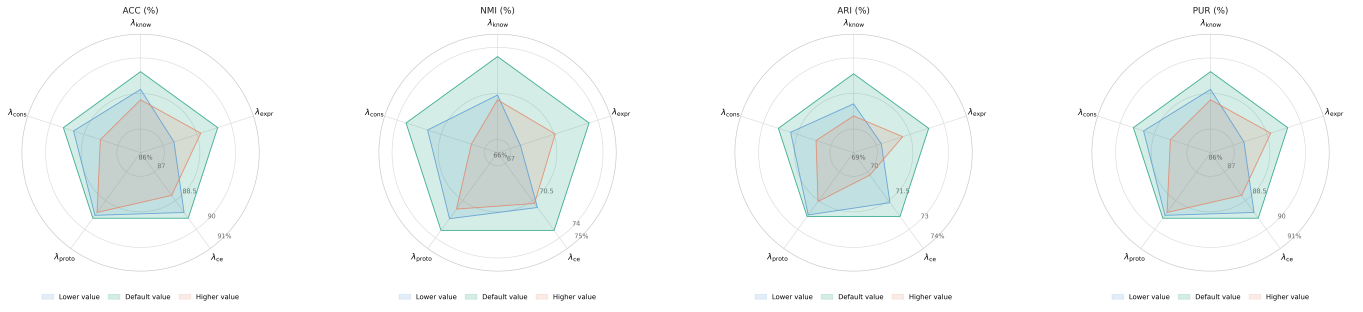


Figure S4 One-factor hyperparameter analysis on BRCA for ACC, NMI, ARI and PUR. Each panel compares lower, default and higher values of λ_{know} , λ_{expr} , λ_{ce} , λ_{proto} and λ_{cons} .

S6. Label-rate analysis

The main manuscript presents the label-rate analysis and its corresponding visualization. Here, we provide the numerical results for the BRCA label-proportion experiment. All profiles were included during training and evaluation, and ACC, NMI, ARI and PUR were computed for the final partitions.

Table S6. Performance of STCKLA under different label availability on BRCA.

Label proportion	ACC	NMI	ARI	PUR
1%	33.23	13.20	6.63	56.63
10%	74.81	55.49	43.55	74.81
20%	89.42	73.30	72.33	89.42
40%	88.38	74.69	68.65	88.38

The numerical results in Table S6 provide the values underlying the main-text label-rate figure.

S7. Implementation and Qwen reproducibility

The repository cited in the main manuscript provides the training, inference, evaluation and plotting scripts, fixed exposed-label splits, configuration files, environment specification, Qwen prompt template and parsed marker-module outputs required to reproduce the reported workflow. The implementation is Python based and uses PyTorch, NumPy, pandas, scikit-learn, matplotlib and tqdm, with CUDA-enabled PyTorch for GPU training. The main benchmark configuration used a 20% exposed-label ratio, seed 0, the number of classes K for k-means, an encoder-visible length of 2500 genes and the loss weights reported in the configuration files.

Qwen3.5-9B was used only for marker-module assembly. The model received target-data candidate genes, evidence summaries from Reactome, TRRUST, GTEx and PanglaoDB, and class context, and returned structured positive markers, negative markers, rejected candidates and rationales. Marker symbols were normalized against the candidate list; duplicated genes and overlaps between positive and negative sets were rejected. Quantitative weights, module activities, model fitting, evaluation and plotting were performed by deterministic scripts or seeded training procedures.

For each candidate gene, the biological-evidence score used in module reliability was

$$s_g = 1.30 n_g^{\text{TRRUST}} + 1.15 n_g^{\text{PanglaoDB}} + 1.10 n_g^{\text{Reactome}} + 1.05 n_g^{\text{GTEx}} + 0.20 n_g^{\text{total}}. \quad (\text{S11})$$

where n_g^{TRRUST} , $n_g^{\text{PanglaoDB}}$, n_g^{Reactome} , n_g^{GTEx} and n_g^{total} denote source-specific and total evidence-hit counts.